

20.

**Ans: A**

magnetic flux density at centre of coil Q,  $B_Q = \frac{\mu_0 N I_Q}{2r} = \frac{\mu_0 I_Q}{2r}$

magnetic flux density at a distance  $2r$  from wire P at centre of coil Q,  $B_P = \frac{\mu_0 I_P}{2\pi d} = \frac{\mu_0 I_P}{2\pi(2r)}$

For the resultant magnetic field to be zero at the centre of coil Q,  
 $B_Q$  and  $B_P$  at the centre must be equal in magnitude and opposite in direction.

$$B_Q = B_P$$

$$\frac{\mu_0 I_Q}{2r} = \frac{\mu_0 I_P}{2\pi(2r)}$$

$$I_Q = \frac{I_P}{2\pi}$$

$$I_P = 2\pi I_Q$$

Since  $B_Q$  is out of the page at the centre of coil Q,  $B_P$  must be into the page at the centre.  
 Hence direction of current in wire P flows to the right.

21.

**Ans: A**